

Data Communication

Content 08: Multiple Access

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Access Control

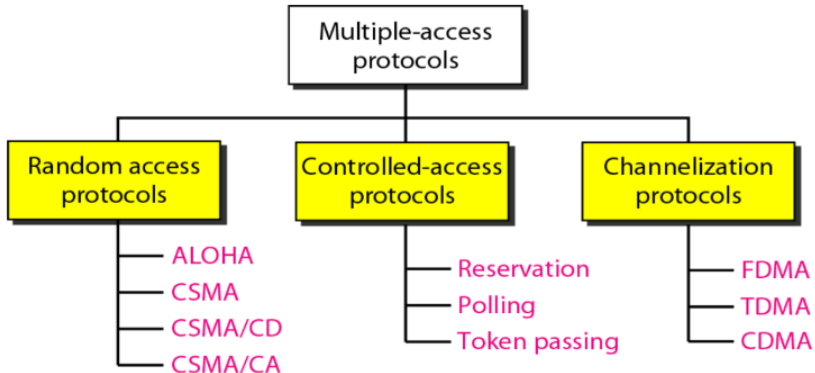
Data Link layer divided into two sub-layers:

- The upper sub-layer is responsible for data-link control
- The lower sub-layer is responsible for resolving access to the shared media



Multiple Access Protocols

When nodes or stations are connected and use a common link, called a multipoint or broadcast link, a multiple-access protocol is needed to coordinate access to the link.



Random Access

- In random access or contention methods, no station is superior to another station and none is assigned the control over another.
- If more than one station tries to send, there is an access conflict – COLLISION – and the frames will be either destroyed or modified.
- To avoid access conflict, each station follows a procedure. The procedure includes instructions on:
 - When can the station access the medium?
 - What can the station do if the medium is busy?
 - How can the station determine the success or failure of the transmission?
 - What can the station do if there is an access conflict?

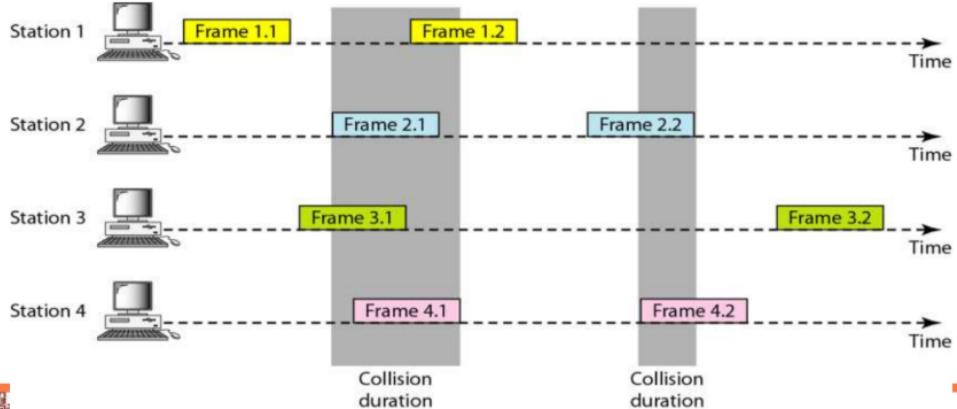


Pure ALOHA

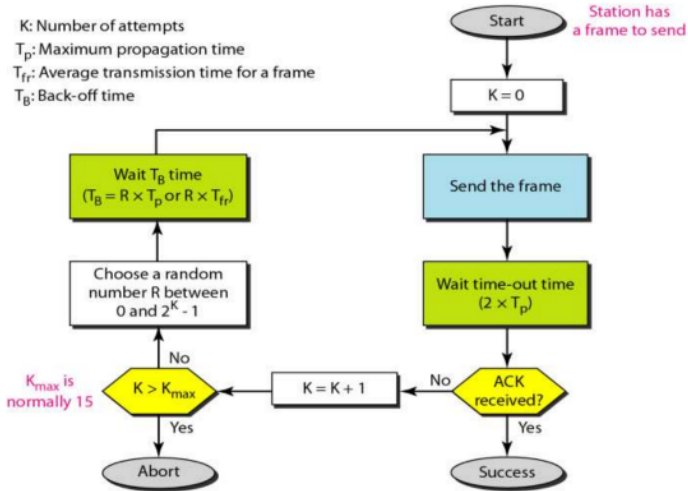
- Each station sends a frame whenever it has a frames to send.
- However, there is only one channel to share, there is the possibility of collision between frames from different stations.



Pure ALOHA



Pure ALOHA: Flowchart



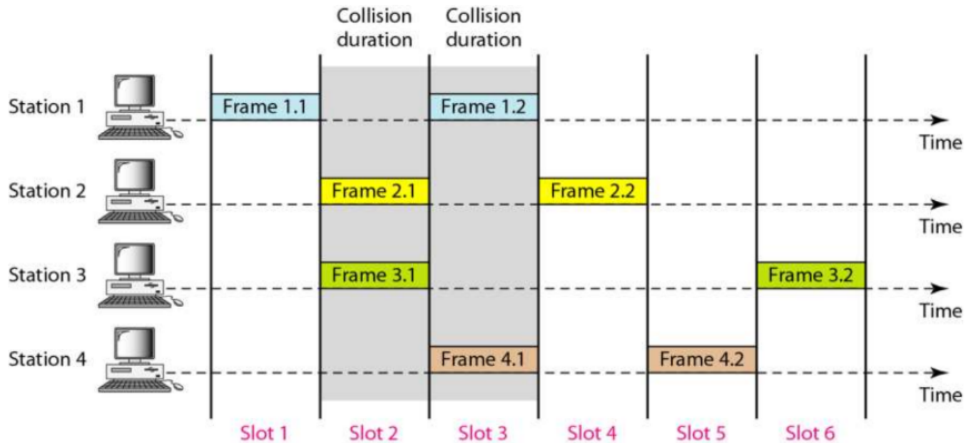
ALOHA Protocol Rule

- Each station sends a frame whenever it has a frames to send.
- After sending the frame, the station waits for an acknowledgment.
- If it does not receive an acknowledgement during the 2 times the maximum propagation delay ($2 \times T_p$),
- It assumes that the frame is lost; it tries sending again after a random amount of time ($T_B = R \times T_P$ or $R \times T_{fr}$)
 - T_p (Maximum propagation time)= distance / propagation speed
 - T_B (Back off time) : Random exponential back off time where R is random number choose between 0 to $2^K - 1$ and K is the number of attempted unsuccessful transmissions
 - T_{fr} (the average time required to send out a frame)



Slotted ALOHA

We divide the time into slots of T_{fr} and force the station to send only at the beginning of the time slot.



Carrier Sense Multiple Access (CSMA)

- CSMA is based on the principle “sense before transmit” or “listen before talk”
- CSMA can reduce the possibility of collision, but it cannot eliminate it.
- The possibility of collision still exists because of propagation delay; a station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.



Persistence Methods

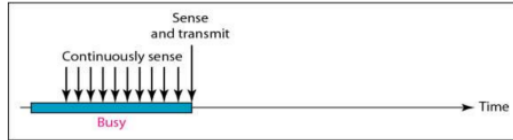
Persistence strategy defines the procedures for a station that senses a busy medium.

Three strategies have been developed:

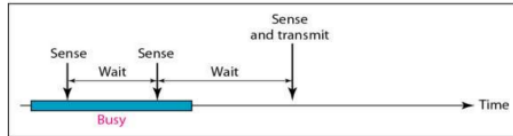
- **1-persistent Method:** After the station finds the line idle, it sends its frame immediately (with probability 1).
- **Non-persistent Method:** A station that has a frame to send senses the line. If the line is idle, it sends immediately. Otherwise, it waits a random amount of time and then senses the line again.
- **p-persistent Method:** After the station finds the line idle, with probability P , the station sends its frames. With probability $q=1-p$, the station waits for the beginning of the next time slot and checks the line again.



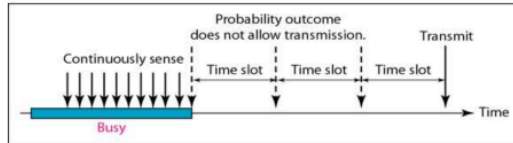
Persistence Methods



a. 1-persistent

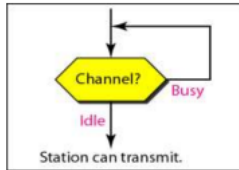


b. Nonpersistent

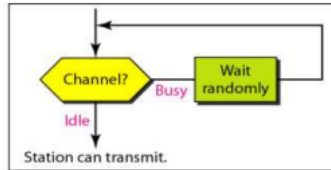


c. p-persistent

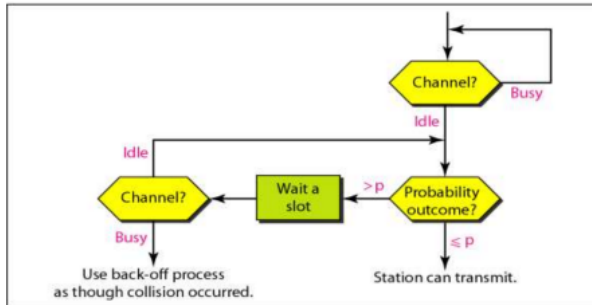
Persistence Methods



a. 1-persistent



b. Nonpersistent



c. p-persistent

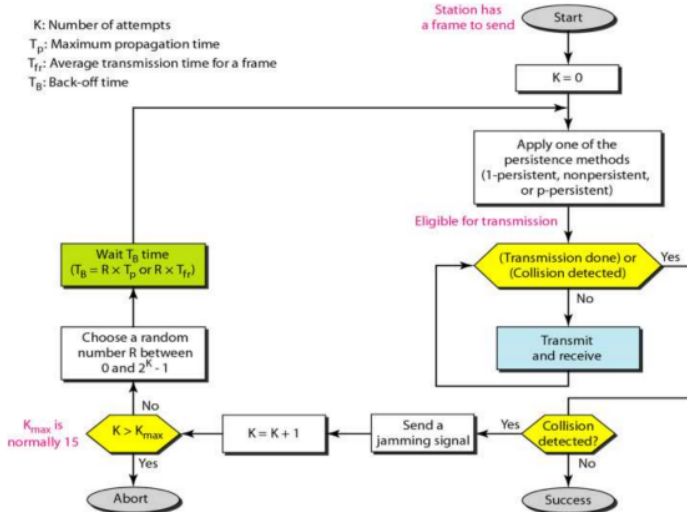
CSMA with Collision Detection (CSMA/CD)

CSMA/CD adds a procedure to handle the collision.

- A station monitors the medium after it sends a frame to see if the transmission was successful.
- If there is a collision, the frame is sent again.



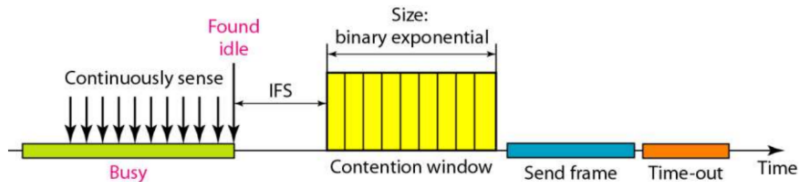
CSMA with Collision Detection (CSMA/CD)



CSMA with Collision Avoidance (CSMA/CA)

CSMA/CA was invented for the wireless networks to avoid collisions. Collisions are avoided through the use of CSMA/CA's three strategies:

- Interframe Space (IFS)
- Contention Window
- Acknowledgements

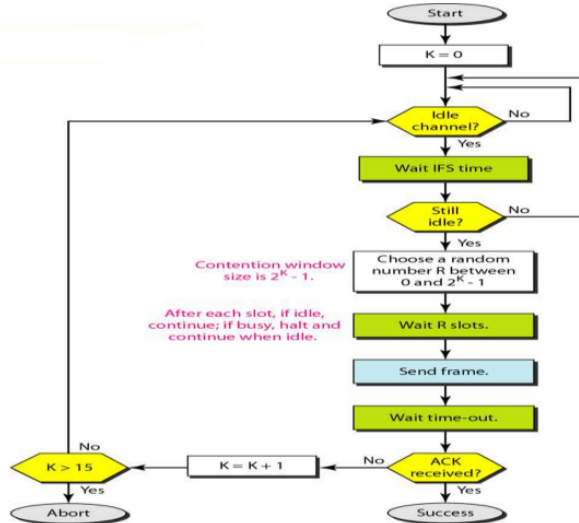


CSMA with Collision Avoidance (CSMA/CA)

- The IFS can also be used to define the priority of a station or a frame.
- If the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

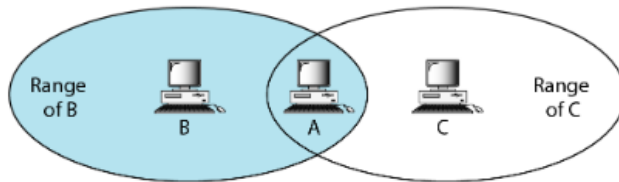


CSMA with Collision Avoidance (CSMA/CA)



Hidden Station Problem

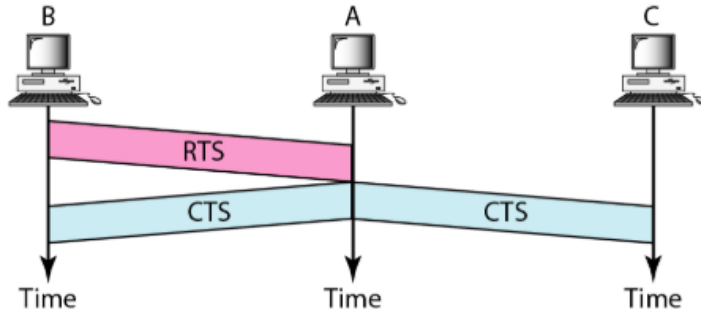
- Station B is sending data to station A.
- Station C also wants to send data to station A.
- Station C is out of B's range and cannot detect B's transmission.
- C assumes the medium is free and starts transmitting.
- A receives transmissions from both B and C, causing a collision.
- Therefore, stations B and C are hidden from each other with respect to A.



B and C are hidden from each other with respect to A.

Hidden Station Problem: Solution

The CTS frame in CSMA/CA handshake can prevent collision from a hidden station.



References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 10: 10.1 - 10.5
Chapter 14: 14.1



Thank You

